

MBA 642: Business Analytics in Healthcare Management

Spring 2026

1/14/2026



Welcome to healthcare analytics!

- This course is one of the courses for the [healthcare analytics concentration](#)
- This course is for everyone
- No previous analytics, statistics, and programming background are required
- All you need is your willingness to learn!



Topics today

1. Course overview
2. Announcement



Topics today

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2. Announcement

What is healthcare analytics?

- Using **information** about patients, treatments, costs, and outcomes to make better decisions
 - Discover patterns in healthcare data
 - Make informed, evidence-based decisions
 - Improve patient care and outcomes
 - Improve systems efficiency (reduce costs while maintaining quality)

Healthcare analytics is useful!

- e.g.,) Hospital Readmission Reduction Program (HRRP) under the Affordable Care Act
 - Zuckerman et al. (2016)

The NEW ENGLAND JOURNAL of MEDICINE

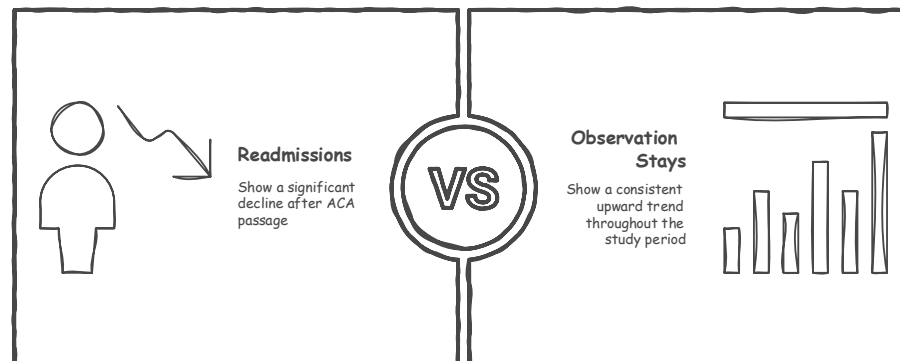
SPECIAL ARTICLE

Readmissions, Observation, and the Hospital Readmissions Reduction Program

Rachael B. Zuckerman, M.P.H., Steven H. Sheingold, Ph.D., E. John Orav, Ph.D.,
Joel Ruhter, M.P.P., M.H.S.A., and Arnold M. Epstein, M.D.

- Actual reduction vs. a “gaming” of the system (stay in observation units)
- Data: Claims from Medicare patients, about 3,300 hospitals, from 2007 to 2015

- Results:



Healthcare analytics is useful!

- e.g.,) [The Wall Street Journal](#) (2014)

In Treatment, There Can Be Too Much of a Good Thing

Aggressive lowering of blood pressure may put patients at risk for harm

By Laura Landro

Aug. 4, 2014 at 7:36 pm ET



Aggressive treatment of chronic diseases such as high blood pressure and diabetes has helped to reduce patient deaths. But a new study suggests some patients may be getting too much treatment, increasing their health risks.

The blood pressure study

Medical records of 400,000 patients who were taking prescription drugs for hypertension

Risk of death or end-stage renal disease: Low > High blood pressure

The blood sugar study

Medical claims from over 33 million beneficiaries

Hospital admission rates: hypoglycemia > hyperglycemia

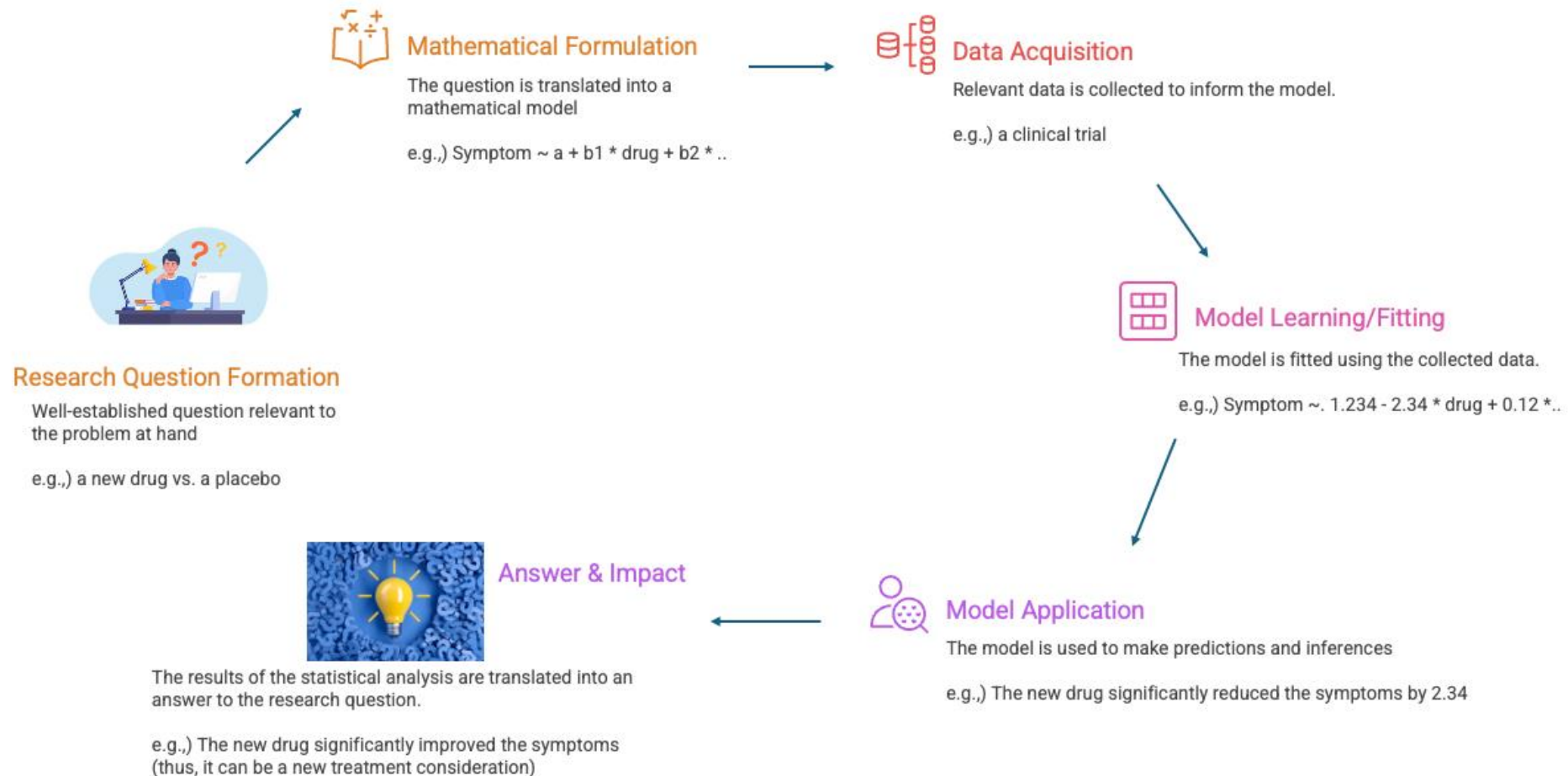


Statistical analysis of data can raise important research questions, leading to the improvement of our health systems

Statistical analysis pipeline

Statistics is not just a model fitting.

It's the entire pipeline **linking questions to answers** through data.

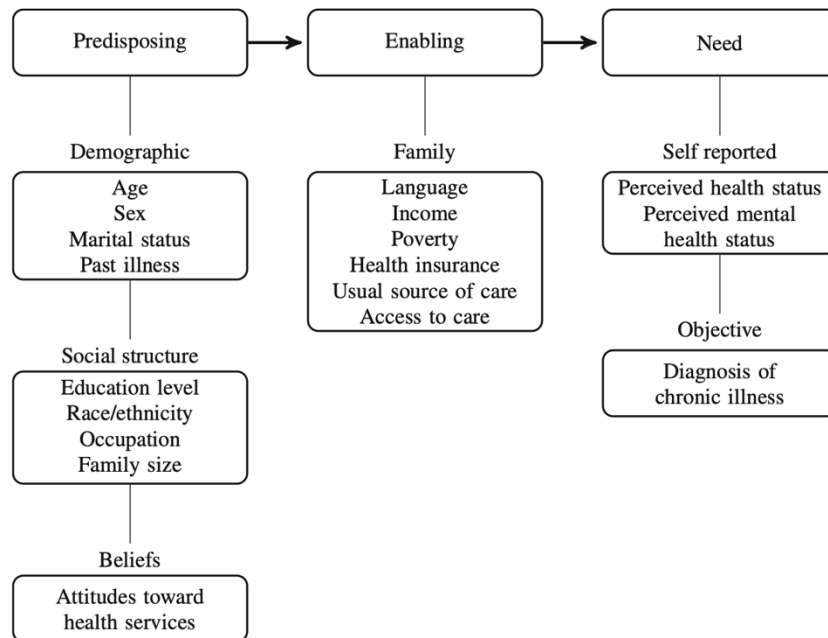


Two approaches to analytics



Hypothesis-driven approach

- Focuses on identifying the underlying **structural phenomenon** (biological, clinical, or socio-economic) that drives heterogeneity in health data
- Uses **statistical inference** and **hypothesis testing** to separate meaningful signals from random noise
- e.g.,) The effect of **physical activity** on **healthcare costs**



The model informs us that

Confounding factors, like *age* or *comorbid conditions*, need to be included to control for their effects on both

Mediating factors, like *BMI* and *self-reported health*, need to be excluded since they can mask the mechanism

Fig. 1.2 The Andersen–Newman model for health services utilization

Predictive approach

- Aims to learn **patterns** in the data to **predict outcomes** with maximum accuracy (e.g., minimizing mean squared error)
- The resulting algorithms are often black boxes, meaning they may not be (mathematically) describable or easily interpreted
- e.g.,) The Heritage Provider Network (HPN) Prize
 - Prediction competition (hospitalization days) using 3 years of claims and demographic data
 - \$3 million prize!
 - Top models used ensemble models (combinations of multiple algorithms)
- Useful to identify **who** is likely to be hospitalized, but hard to explain **why**
- **Kaggle.com** is a place where you can find various competitions and analytic approaches

Clarify your goal: mechanism vs. prediction

- It's not about one being better than the other
- The choice between traditional and predictive paradigms depends on the **objective of the analysis** and the intended use of the results
- Developing understanding and familiarity with both paradigms enables the selection of appropriate methods tailored to diverse research goals

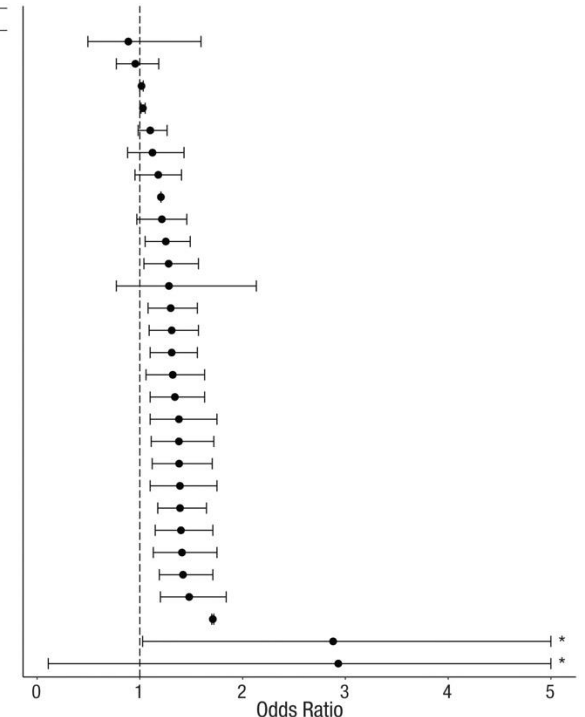
One dataset, multiple valid approaches

- Multiple valid statistical approaches exist for any research question
 - There is no single "correct" analysis!
- Understanding how various tests work helps us evaluate whether a chosen approach is appropriate

e.g.,) Many Analyst Project (Silberzahn et al., 2018)



Team	Analytic Approach	Odds Ratio
12	Zero-Inflated Poisson Regression	0.89
17	Bayesian Logistic Regression	0.96
15	Hierarchical Log-Linear Modelling	1.02
10	Multilevel Regression and Logistic Regression	1.03
18	Hierarchical Bayes Model	1.10
31	Logistic Regression	1.12
1	OLS Regression With Robust Standard Errors, Logistic Regression	1.18
4	Spearman Correlation	1.21
14	WLS Regression With Clustered Standard Errors	1.21
11	Multiple Linear Regression	1.25
30	Clustered Robust Binomial Logistic Regression	1.28
6	Linear Probability Model	1.28
26	Hierarchical Generalized Linear Modeling With Poisson Sampling	1.30
3	Multilevel Logistic Regression Using Bayesian Inference	1.31
23	Mixed-Model Logistic Regression	1.31
16	Hierarchical Poisson Regression	1.32
2	Linear Probability Model, Logistic Regression	1.34
5	Generalized Linear Mixed Models	1.38
24	Multilevel Logistic Regression	1.38
28	Mixed-Effects Logistic Regression	1.38
32	Generalized Linear Models for Binary Data	1.39
8	Negative Binomial Regression With a Log Link	1.39
20	Cross-Classified Multilevel Negative Binomial Model	1.40
13	Poisson Multilevel Modeling	1.41
25	Multilevel Logistic Binomial Regression	1.42
9	Generalized Linear Mixed-Effects Models With a Logit Link	1.48
7	Dirichlet-Process Bayesian Clustering	1.71
21	Tobit Regression	2.88
27	Poisson Regression	2.93



One dataset, multiple valid approaches

- An example in a neuroimaging study

Article

Variability in the analysis of a single neuroimaging dataset by many teams

<https://doi.org/10.1038/s41586-020-2314-9>

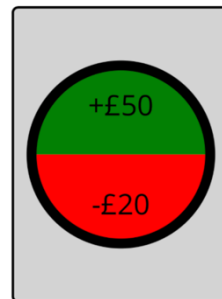
A list of authors and affiliations appears in the online version of the paper.

Received: 14 November 2019

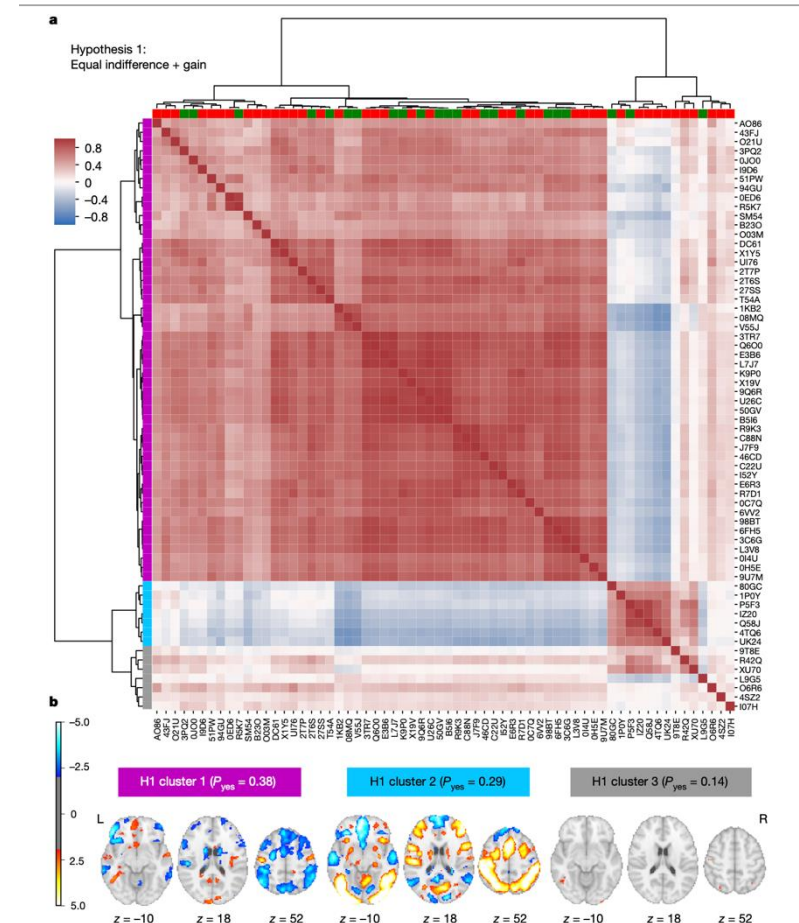
Accepted: 7 April 2020

Published online: 20 May 2020

Data analysis workflows in many scientific domains have become increasingly complex and flexible. Here we assess the effect of this flexibility on the results of functional magnetic resonance imaging by asking 70 independent teams to analyse



Accept vs. Reject



Do we need analytics skills in the LLM era?

DATA VISUALIZATION

I Tested Frontline M-LLMs on Their Chart Interpretation Skills

Can multimodal LLMs infer basic charts accurately?

Thuwarakesh Murallie
Nov 5, 2024 • 35 min read



Chart Type	Challenge	Meta Llama 3.2 11b	Meta Llama 3.2 90b	Open AI GPT-4o	Mistral Pixtral 12b	Claude 3.5 Sonnet	Google Gemini 1.5
Bar chart	Find the most popular fruit and the difference with the least popular.	Green	Green	Green	Green	Green	Green
Line Chart	Find the hottest day of the week and the difference with Sunday	Green	Green	Yellow	Green	Green	Green
Pie Chart	Find the size of a single slice	Green	Yellow	Green	Red	Yellow	Red
Stacked Bar Chart	Find the total value of one segment across the bars	Yellow	Yellow	Yellow	Red	Green	Red
Scatter plot	Estimate the exam score from studying time	Yellow	Red	Green	Yellow	Green	Green
Multiple Line Chart	Find the line with biggest growth	Green	Yellow	Red	Red	Green	Green
Histogram	Find the median age and the % below 30	Red	Red	Green	Green	Green	Green
Box plot	Find the IQR and comment on it	Green	Red	Green	Green	Green	Red

- It's not perfect, so you cannot completely rely on it
- You still need to be able to [verify](#) whether the analysis approach, processes, and results are accurate
- LLMs are tools, not replacements for understanding!

Topics in this course

- Basics of R and R-Studio (Week 2)
- Basic data analysis (Week 3 – Week 4)
 - How to describe and visualize data
- Mean comparisons (Week 5 – Week 7)
 - t-test and ANOVA
- Association between variables (Week 8)
 - Crosstab and correlation
- Prediction models (Week 11 – Week 16)
 - Linear regression, logistic regression, conjoint analysis, penalized regression, and tree models

Course Objectives

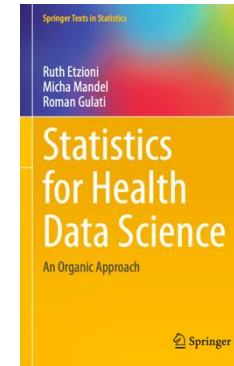
- By the end of the course, students will be able to:
 - Understand the role and application of analytics in healthcare management decision-making
 - Conduct exploratory data analysis on different types of healthcare data (patient surveys, health outcomes, etc.)
 - Select and apply appropriate statistical methods to healthcare management problems to extract meaningful insights
 - Implement these analytic skills in R
 - Interpret analytical results and communicate findings effectively to diverse stakeholders
 - Design and implement data-driven solutions for real-world healthcare management challenges

Course Prerequisite

- There are no prerequisites for this course
- This course is designed to accommodate students with varying levels of statistical and programming backgrounds
- All necessary concepts will be introduced from the ground up

Textbooks

- Two textbooks (recommended) will be mainly used for this course:
 - **Statistics for Health Data Science: An Organic Approach**
 - Authors: Ruth Etzioni, Micha Mandel, and Roman Gulati
 - ISBN: 978-3-030-59888-4
 - **Statistical Analytics for Health Data Science with SAS and R**
 - Authors: Jeffrey Willson, Ding-Geng Chen, and Karl E. Peace
 - ISBN: 978-1-032-32569-9
 - Both books are freely available through the UD library



Other helpful books and readings

- There are also many freely available, useful learning resources.
 - R for Data Science
 - Authors: Hadley Wickham and Garrett Grolemund
 - Introduction to Modern Statistics
 - Authors: Mine Çetinkaya-Rundel and Jeffrey S. Simonoff
 - Statistical Thinking for the 21st Century
 - Authors: Russell A. Poldrack

Required Software: R and R-Studio

- This course utilizes **R** and **R-Studio** as the primary analytical tools.
 - **R** is a powerful, open-source programming language specifically designed for statistical data analysis, widely used in both academia and industry for analytics
 - **R-Studio** is an integrated development environment (IDE) that makes R programming more accessible and efficient through its user-friendly interface, offering features such as data visualization, code editing, debugging, and project management capabilities
 - Students will learn to use these tools to analyze healthcare data, create visualizations, and develop and fit statistical models

Required Software: R and R-Studio

- We will use R and R-Studio in every class starting next class, so make sure you have them ready before the next class
 - Download R: <https://cloud.r-project.org>
 - Download R-Studio: <https://posit.co/downloads/>
 - Let's take some time and install them. You first need to install R

Grading components

Component	Points
Assignments	90 points
Class Participation and Reflection	78 points
Midterm Exam	50 points
Final Exam	50 points
Total	268 points

1. Assignments (90 points)

- There will be 9 assignments throughout the semester (10 points each)
- Each assignment requires students to apply the analytical methods learned in the class session to new healthcare data analytics scenarios
 - e.g.,) In Week 4, after we cover data visualization, you will be asked to visualize results with a new healthcare dataset
- Assignments will typically take about 30-60 minutes to complete
- Assignments are due by Sunday at 11:59 PM of the week

2. Class attendance, participation, and reflection (78 points)

- After we finish each session, I would like to hear about what you learned and what questions remain unclear
- To encourage engagement with the course contents, you will need to write a short list of responses to the following questions
 - What were the three most interesting or exciting things you learned from the session? And why? (3 points; 1 point for each one)
 - What were the three most unclear or confusing things from the session? And why? (3 points; 1 point for each one)
- This one starts next week (no reflection assignment this week)

2. Class attendance, participation, and reflection (78 points)

- You don't need to get stressed about this
 - I want to understand what genuinely resonated with you or what felt challenging
 - **For example:**
 - *“Learning how to draw a bar graph using ggplot was interesting. I didn't know how to draw graphs in R before, so discovering this was exciting.”*
 - “Understanding the interaction effect was confusing. It's unclear to me whether it means the effect of A is different from the effect of B, or whether the effect of A changes depending on B.”
- These kinds of honest, straightforward reflections are exactly what I'm looking for
 - 2-3 sentences for each point are enough

3. Mid-term exam (50 points)

- In Week 9
- Covers topics from Week 2 through Week 9
 - Descriptive statistics, data visualization, t-test, ANOVA, crosstab, and correlation
- The format will be similar to assignments
- You will be given the whole day (from 12 AM to 11:59 PM) to complete it

4. Final exam (50 points)

- In Week 17 (Final Exam week)
- Covers topics from Week 11 through Week 16
 - Linear regression, logistic regression, conjoint analysis, penalized regression, and decision trees
- The format will be similar to assignments and the mid-term exam
- You will be given the whole day (from 12 AM to 11:59 PM) to complete it

Late submissions

- Life gets busy, and I understand that sometimes things don't go as planned. However, please try your best to submit on time. Here's what happens if you're late:
 - **Reflections**: 1.5 points deducted for every 24 hours late. Submissions over 48 hours late receive zero points.
 - **Assignments**: 2 points deducted for every 24 hours late. Submissions over 48 hours late receive zero points.
 - **Exams**: Must be completed within the specified time frame. No late submissions accepted.

Final Course Letter Grades

Final course letter grades will roughly be based on the following criteria:

Letter Grade	Quality Points	Point Range
A	4.000	250 - 268
A-	3.667	240 - 249
B+	3.333	230 - 239
B	3.000	220 - 229
B-	2.667	210 - 219
C	2.000	185 - 209
F	0.000	0 - 184

Any decimal numbers will be rounded to the nearest integer

Important class policies

- **Camera on**, please! Keeping your camera on helps create a more engaging and connected learning environment.
- **Attendance:**
 - Regular attendance is expected
 - Attendance will be checked at a random point during each session, or reflection assignments will be used
 - If you must miss a class, please let me know in advance when possible. All lecture recordings, materials, and announcements will be available on Canvas so you can catch up.
 - Missing class does not relieve your responsibility for assignments, reflections, or exams, as these are due on set deadlines regardless of attendance.

Office hours

- **Hours** – Wed 1:30 PM – 3:00 PM or after class or by appointment
- **Office** – Zoom (check the link on Canvas)
- **Contact** – email: syoon1@udayton.edu
- Preferred form of contact is **Email**
 - Please [specify your class](#) (MBA642) or time in the email
 - I'll try my best to respond to you within 24 hours (excluding weekends and holidays)

Canvas and class materials

- All class materials and announcements will be posted on Canvas
 - Class slides or link will be posted before or after each class
 - Additionally, I'm using my [GitHub repository](#) and [page](#) for managing data and scripts
 - Check the Canvas page frequently, and do not ignore the class slides if you missed the class

Intellectual property

- All course materials (e.g., assignments, examinations, worksheets, problems, projects, documents, recordings, or other materials) are protected
- They cannot be reproduced, distributed, or transmitted via scanning, photographing, copying, uploading, or other electronic methods without permission

Final comments

- All students will be treated fairly and respectfully, regardless of background (ethnicity, nationality, religion, gender, etc.). Grades will be based solely on performance in terms of requirements listed in the syllabus
- If you are ever confused or struggling, please come to office hours or reach out to me. It is my job to support your learning experience at UD. Please don't keep your struggle to yourself



Topics today

1. Course overview
2. Announcement

Next class (01/21)

- We will have an intro session to R, R-Studio, and data wrangling
 - Please make sure **R and R-Studio** work before coming to the next class